

INDIA’S AI STACK

				
LAYER	WHAT IT IS	CURRENT STATUS	KEY DATA & EXAMPLES	RISKS/OPPORTUNITIES
LAYER 1 Hardware base	GPUs, HBM, Fabs, EDA (electronic design automation) tools	Weak—High import dependence; yet to get local fabs	• 34,000 GPUs procured (IndiaAI Mission) • No advanced-node fabs • EDA tools dominated by US firms	✗ Strategic dependence ✓ Opportunity to invest in sovereign silicon ecosystem
LAYER 2 Model layer	LLMs & SLMs	Emerging—Late to LLM race, SLMs gaining traction	• Missed GPT-level frontier models • Indian SLMs: Sarvam AI, Kissan AI, Krutrim, Infosys’ Project Indus	✗ Lag in foundational models ✓ Edge in cost-efficient, contextual SLMs for India’s diverse markets
LAYER 3 Application layer	Enterprise use, DPI integration, Startups	Growing—DPI backbone strong, enterprise adoption rising	• UPI, Aadhaar, ONDC as AI deployment platforms • 5x ROI from GenAI (Microsoft research for enterprises) • 100+ GenAI start-ups emerging	✗ Leverage DPI + AI for productivity at scale ✓ Limited AI penetration in Tier-2/3 enterprise segments
LAYER 4 Outcome layer	Economic returns, productivity, digital sovereignty	Ambitious—Goal set, but structural gaps persist	• Target: \$1T AI economy by 2030 • 256% AI talent growth (since 2016) • 18M developers (2nd after US)	✗ May remain potential unless full stack is built ✓ India can leapfrog by aligning DPI + AI + SLMs at population scale

AI power users in the country now begin their day with AI. This is a shift in how work gets done, how decisions are made, and how value is created. India’s strength lies in its people.”

India cannot afford to rely solely on foreign AI systems, experts warn. Dr Vivek Raghavan, co-founder of AI venture Sarvam, argues that such dependence is strategically short-sighted. “India has all the ingredients to lead: linguistic and cultural diversity, a fast-digitising economy, ambitious digital public infrastructure,

and deep technical talent. What we need now is the will to align these strengths into a long-term national capability,” he says.

“If India had its own cloud, we’d control our data. Without it, we’re dependent,” says Punit Pandey, founder and Chief Information Officer of Ojas Softech and Astrosage AI. “Some say, ‘We’re doing fine without it,’ but how? We have no global tech products despite having 30–50% of Silicon Valley’s engineers. This isn’t ‘fine’. Colonial history may have made us risk-averse, but we must change.”

Only by securing the full Silicon value chain can India safeguard its AI growth trajectory, reduce strategic vulnerabilities, and transform from a global back-office to an end-to-end AI powerhouse, say experts.

At the end of the day, the question remains: Can India’s bold AI ambition translate into real economic power or will it remain just another unfinished chapter in its long saga of potential waiting to be tapped. **BT**

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